PROJECT SAATHI: TECHNICAL EVALUATION REPORT

Offline-First AI Clinical Screening & Autonomous Referral Infrastructure for Rural Healthcare

Document Purpose: Official Project Attestation for Competition Cost Reimbursement | **Track:** Assistive Healthcare AI & Distributed Cloud | **Event:** Build 2026

1. EXECUTIVE OVERVIEW & RURAL HEALTHCARE BOTTLENECK

In rural India, over 10 lakh Accredited Social Health Activist (ASHA) workers are the foundational point of primary healthcare. However, frontline operations suffer from three systemic barriers: **unreliable 2G or zero network connectivity**, **regional language illiteracy**, and an absence of localized specialist diagnostic support. Consequently, early symptoms of malignant skin lesions, diabetic retinopathy, respiratory infections, and hematological abnormalities are missed, leading to critical late-stage presentations at secondary hospitals.

Project Saathi resolves this paradigm by transforming commodity frontline smartphones into intelligent, edge-computed clinical assistive screening hubs. Operating 100% offline at the doorstep, it establishes an unbroken continuum from community screening to Community Health Centre (CHC) clinician validation, while synchronizing aggregated epidemiological insights to the District Medical Officer (DMO) command dashboard.

2. SYSTEM ARCHITECTURE & TECHNICAL INNOVATIONS

- **Edge-AI Multi-Pathway Triage (100% Offline Resilience):** Employs quantized on-device computer vision models running across 4 clinical pathways: dermatological skin lesions, optical fundus diabetic retinopathy, digital chest radiographs, and laboratory CBC hematology parameters. Real-time visual quality gates validate blur and luminance before scoring, instantly triaging risk into non-diagnostic traffic-light states (Green, Amber, Red). All patient journeys, consent states, and image metadata persist on an ACID-compliant edge SQLite store with zero internet dependence.
- **22 Constitutional Indian Languages Voice Engine:** Implements full audio guidance and conversational voice synthesis across all 22 official scheduled Indian languages. ASHA workers receive audible spoken task briefings on pending patient visits and audible guidance during camera capture, eliminating literacy friction.
- **Cloud Autonomous Agent for Hospital OPD Token Booking:** Integrates autonomous web AI browser agents via cloud-orchestrated infrastructure. When an at-risk (Red/Amber) flag is identified, the agent autonomously navigates public hospital e-Hospital portals to pre-book referral OPD appointments, generating digital queue tokens that save rural patients hours of physical transit and triage delay.
- **Render Workflows Distributed DAG Pipeline (@renderinc/sdk/workflows):** Employs Render's distributed orchestration engine to execute durable multi-step data pipelines upon field synchronization: (1) Ingestion and cryptographic hash validation of offline sync batches, (2) Strict enforcement of India's **Digital Personal Data Protection (DPDP) Act 2023** via automated Small-Cell Suppression (< 5 cases masked to prevent patient re-identification), (3) Dispatch of autonomous CHC referral tokens, and (4) Append-only cryptographic ledger generation for medical-legal accountability.

3. TECHNICAL JUSTIFICATION OF PROJECT REIMBURSEMENT EXPENSES

The claimed financial expenditure directly reflects infrastructural compute, live cloud orchestration, specialized multi-lingual AI models, and physical testing accessories required to build and benchmark a genuine, operational full-stack prototype:

Category & Resource	Technical Component & Service	Engineering Purpose & Necessity in Project Saathi
Distributed Cloud Compute	Render Workflows & Cloud Web Services	Durable multi-task DAG orchestration for DPDP privacy masking, background sync resolution, and Express REST API.
Voice AI Synthesis	ElevenLabs Multilingual V2 Speech API	High-fidelity natural clinical speech generation across 22 Indic regional languages with on-device phonetic fallback.
Autonomous Web Agent	Cloud Browser-Use Execution Engine	Headless virtual browser infrastructure executing automated OPD token pre-booking on public hospital portals.
Vision & Multimodal AI	Google Gemini Multimodal API	High-resolution lesion texture validation, optical quality scoring, and clinical safety counter-checking.
Edge Testing Hardware	Camera Rigs, Optical Couplers & Devices	Smartphone test rigs, physical lens attachments for fundus simulation, and offline network throttling apparatus.

4. LIVE PRODUCTION DELIVERABLES & VERIFICATION URLS

ASHA & Doctor Mobile App:	https://project-saathi-rho.vercel.app/app	**District Admin Portal:**	https://project-saathi-rho.vercel.app
Render Workflows Live API:	https://project-saathi-api.onrender.com/health	**Open Source Codebase:**	https://github.com/VanshikaB16/project_saathi

5. ATTESTATION & COMPLIANCE STATEMENT

This project was conceptualized, architected, and built from scratch adhering strictly to the highest medical-legal standards: screening flags are strictly non-diagnostic assistive triage; synthetic demo records are strictly partitioned from real data; and patient anonymity is safeguarded via non-biometric municipal tokens and automated Small-Cell Suppression. All claimed expenses correspond directly to functioning technical infrastructure verified through the live endpoints above.